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Comments on Amerindian Hunting Practices in Trinidad (West Indies): Tetrapods From the Manzanilla Site (Late Ceramic Age 300–900 AD)

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ABSTRACT

Located on the eastern shore of Trinidad island, the Manzanilla (SAN-1) deposit is a lengthy pre-Columbian occupation sequence consisting of an extensive Palo Seco (Saladoid) layer (300–600 AD) and a Bontour layer (600–1400 AD). According to initial reports on faunal remains, the former inhabitants of this site seem to have developed acquisition strategies relying on a wide range of ecological niches, including marine and terrestrial resources. A wide faunal spectrum is present in the Manzanilla assemblage, illustrating an opportunistic exploitation of the local surroundings. Marine resources, such as mollusks and fish, have indeed played a continuous major role in pre-Columbian diet, but the significant amount of tetrapod remains show that terrestrial game was also significant. Published zooarchaeological material is relatively scarce in this region, particularly in Trinidad. Therefore, through the quantitative and qualitative analysis of the faunal remains (skeletal parts distribution, ungulates osteometry) this study aims to shed some light on the hunting practice patterns and food harvesting strategies of the former inhabitants.

Keywords Caribbean, economy and subsistence, zooarchaeology

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INTRODUCTION

As a continental island located near the South American mainland (Figure 1), Trinidad presents a wide diversity of terrestrial fauna, unlike other islands in the West Indies. The smaller sister island of Tobago lies 32 km northeast and shares some of its biogeographical features, including the presence of some mainland species not found elsewhere in the Caribbean (Steadman and Jones 2006; Steadman and Stokes 2002). As the first West Indies island colonized by pre-Columbians from the Continent (Boomert 2000), Trinidad is an interesting place to study the Amerindian cultures from the Ceramic Age. Though the Late Ceramic age settlers relied strongly on both marine and terrestrial resources (Boomert 2000), we choose to focus our study on land animal remains to obtain a more accurate view on the hunting practices.

Between 1997 and 2007, an international team of archaeologists from Trinidad and the Netherlands carried out several excavations on the Manzanilla plateau, in eastern Trinidad. These fieldwork operations renewed our knowledge of pre-Columbian settlements on the eastern shore of the island (Boomert 2000; Nieweg and Dorst 2001). However, with the exception of a preliminary study in the 1998 excavation report, partially published in the archaeological and ethnohistorical synthesis about Trinidad and Tobago by Boomert (2000), no original investigations on faunal remains have yet been published. The aim of this study, therefore, is to shed new light on Amerindian subsistence strategies in Trinidad through the analysis of faunal assemblages from a Late Ceramic Age site. In this paper, we will focus primarily on terrestrial food sources based on hunting practices from a wide spectrum of environments.

METHODS

Assemblage Collection

The faunal assemblage presented here was uncovered during the 1997 archaeological survey campaign. Three test pits were

dug in different parts of the midden in order to evaluate the stratigraphy and limits of the Amerindian settlement. The excavation began with the digging of arbitrary levels of up to 10 cm. The sediments were dry sieved through 1/4-inch mesh and the material recorded within each of these layers. In units 1 and 2, half of the soil was water sieved using a 1/6-inch screen (Nieweg and Dorst 2001).

These 10-cm levels were then aligned with the cultural layers using the field drawings of the stratigraphic section of Unit-1 (Nieweg and Dorst 2001) (Figure 2). Schematically, three main stratigraphic contexts were observed in Unit-1 and referenced as 1 (top), 2 (middle), and 3 (bottom).

Analytical Methods and Tools

The taxonomic identifications were realized using different osteological collections of the “Caribe”: “Antilles bone collection” in UMR 7209, the *Laboratoire d'Anatomie Comparée et the Mammifères et Oiseaux* collections from the *Muséum National d'Histoire Naturelle* in Paris. We identified up to species level where possible; however, in most cases only the genus or the family rank could be identified due to the high degree of fragmentation of some specimens.

Three units of quantification were used to evaluate the relative abundance of each taxa: NISP, MNI, and Bone Mass (BM). NISP and MNI are units commonly employed in zooarchaeology to quantify taxonomic abundance with varying degrees of benefits and limitations, as emphasized in the profuse literature dealing with this issue (Grayson 1984; Klein and Cruz-Uribe 1984). For example, while NISP is the most direct way to quantify a faunal collection, its estimation can be profoundly affected by specimen fragmentation (Cannon 2013). Conversely, MNI is less biased by fragmentation but its calculation is more complicated and widely depends on sample size and assemblage aggregation techniques (Grayson 1984; Klein and Cruz-Uribe 1984). For this study, the MNI was calculated using the most frequent element of each taxa from each cultural layer, offset against the relative parameters (age,

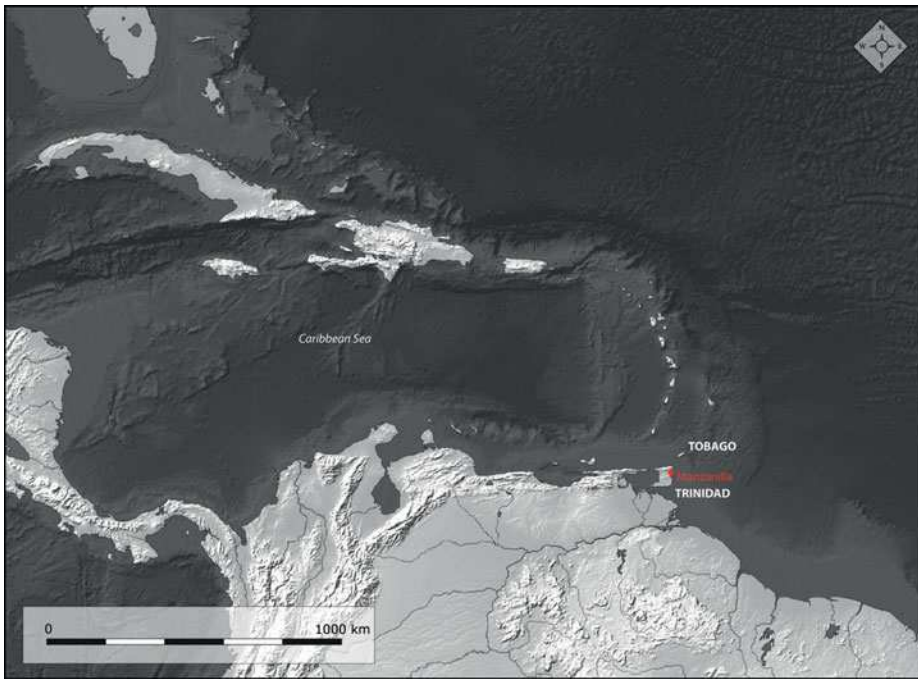


Figure 1. Location of Trinidad in the West Indies. Manzanilla site is located on the eastern coast of the island.

size). The total MNI presented in this article is the aggregate of these calculations.

These results serve as the basis for our analysis, including an estimation of faunal richness and diversity and their chronological evolution. In order to infer the specific richness and diversity of the Manzanilla assemblage, we calculated the Margalef richness index¹ (Bobrowsky and Ball 1989; Margalef 1958) and the inverse of Simpson index² (Begon et al. 2006).

In order to assess the relative contribution of the different ecosystems, the edible biomass of the different taxa has been evaluated using an allometric formula based on the specimen's weight, as proposed by Reitz and Wing³ (2008:236–241).

Cervids, peccaries, and caviomorphs were deemed the most suitable for skeletal distribution analysis⁴ (Behrensmeyer and Dechant Boaz 1980; Dodson and Wexlar 1979; Grouard 2001), as they were the most common mammals from the deposit. To complete this approach on carcass process-

ing we also used another method, mainly based on Binford's works on big game rendering and transport strategies (Binford 1978; Faith and Gordon 2007). Faith and Gordon's (2007) approach permits the association of a specific osseous subset with one of the hypothetical carcass management strategies, as described by Binford. These calculations were applied to the two artiodactyl species identified in the assemblage (i.e., brocket deer and peccary).

Among the entire set of Manzanilla tetrapod bones, artiodactyls were one of the most numerous and best preserved. Some of the bones were osteologically complete, allowing measurements to be taken according to von den Driesch's standards (Driesch 1976). When analyzing the collection, we systematically recorded all metrical data where possible, even on partially complete specimens. We then compared them to measurements recorded on modern cervids and tayassuid specimens from the osteological collections of the *Muséum national*

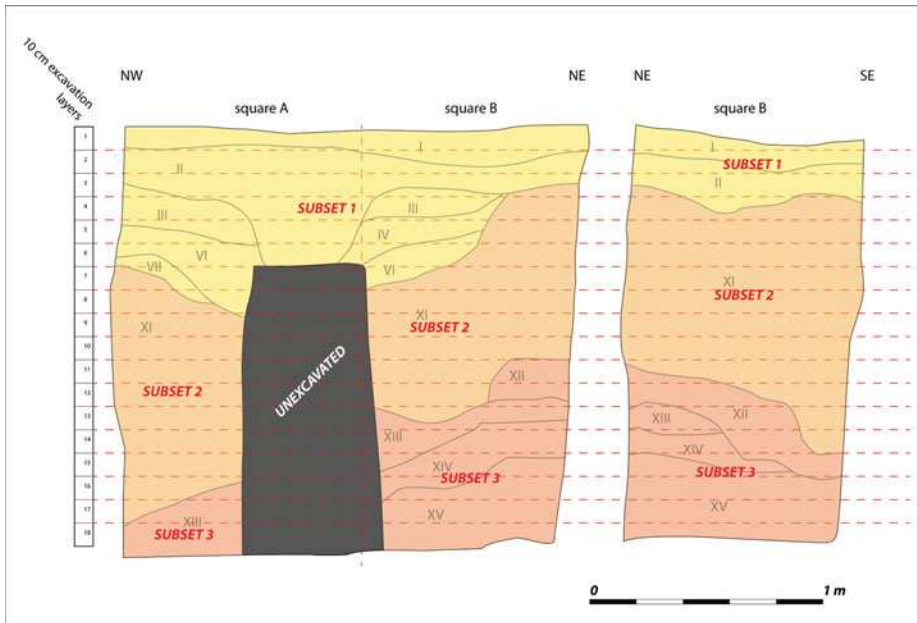


Figure 2. Section of test pit No. 1 (theoretical 10-cm layers are drawn in red) (figure modified from Nieweg and Dorst 2001).

d'Histoire naturelle. Deer bones came from two *Mazama americana* individuals⁵ (male and female) from the museum zoo. Captivity may have some morphological consequences on the structure and the size of wild animal bones like a decrease of osseous robustness or an enlargement of long bones (O'Regan and Kitchener 2005). It may thus be a biasing factor in comparative analysis, especially when studying osteometry. Measurements from two other cervid individuals of the *Odocoileus virginianus* species were also recorded for our study.⁶ Regarding peccaries, the comparison set is composed of four individuals of the *Pecari tajacu* species.⁷

BACKGROUND

Geographical and Archaeological Context

The island of Trinidad (4,827 km²) is the fifth largest Caribbean island, located 15 kilometers from the Venezuelan coast. As the

island lies on the South American Continental Shelf, the Orinoco and the South Equatorial Current from the Amazon directly influence it. Consequently, the biota and terrestrial habitat of Trinidad have more affinities with the ecology of equatorial South America than with the other Caribbean islands, except Tobago (Eisenberg 1989; French 1991; Pregill et al. 1994).

Most of the surface formations of Trinidad date from the Cretaceous and Cenozoic, although some rocks can be dated back to the Jurassic. The Northern range, which is the easternmost extension of the Venezuelan Cordillera, is formed of metamorphic rocks uplifted during the Jurassic and Cretaceous periods (Kugler 2001), while the Central and Southern ranges are composed of Tertiary rocks. Separation from the continental mainland occurred in more recent geological times, about 11,000 years ago for Tobago and 5,000 years ago for Trinidad. The Holocene sea level was not homogeneous (Figure 3), with a rapid rise (5 mm/yr) occurring between 8,000 and 6,000 BP, a low rate



Figure 3. Location of Manzanilla-1 site on the island of Trinidad (© Google maps). According to Wing (1962), the 24 fathom line is considered to be the shoreline during late Pleistocene (c. 18,000 BP) and the 12 fathom line the shoreline of the early Holocene era.

of increase (0.25 mm/yr) after 4,000 BP and an intermediate rate from 6,000 to 4,000 BP (Ramcharan 2004).

The surface of the island can be divided into five main geographical regions (Figure 4). Three mountain ranges cross it from east to west. The Northern Range is the highest mountainous area of the island, with a maximum elevation of 940 meters (*Cerro Aripo*), stretching from the northern coast to the Caroni River valley. Southwards the other two mountainous ranges, the Central and Southern Range, are systems of moderate altitude hills. The Caroni, Oropuche, and Ortoire rivers drain the lowlands located between these mountains (Beard 1946; Boomert 2000). Extensive marshlands can also be found along the coastal plains. The Manzanilla shell midden lies just between the large freshwater marsh of the Nariva Swamp and the Central Range, on the eastern shores of Trinidad.

During pre-Columbian times, the unique position of Trinidad made the island an im-

portant cultural hub between the Orinoco basin, the Guyana plateau, and the Caribbean archipelago. Indeed, the Ortoiroid Amerindians (Mesoindians, Archaic Age) first occupied Trinidad between about 6,000 and 4,000 cal BC, when the Gulf of Paria and the Columbus Channel gradually attained their present level and configuration (Boomert 2009). Simultaneously, they inhabited several islands in the Lesser and the Greater Antilles (Bonnissent 2010). Around 300 cal BC, the first groups of pottery-making horticulturalists, called Saladoid Amerindians (Ceramic Age, Neoindians), settled in Trinidad from the eastern littoral of Venezuela, the Orinoco delta, and the coast of Guyana (Boomert 2009) and eventually colonized the Lesser Antilles archipelago as far as Puerto Rico.

The Manzanilla (SAN-1) Site. The Manzanilla (SAN-1) site is located on the top of a 15-meter-high hill, about 200 meters eastwards of Cocos Bay, on the eastern shores of Trinidad. This small plateau dominates

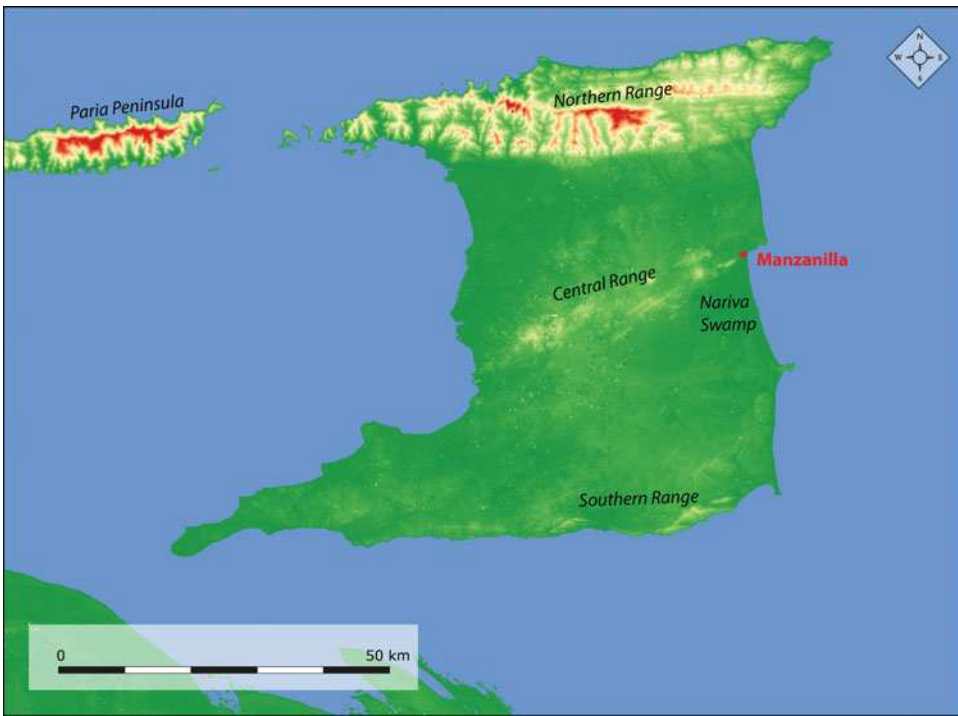


Figure 4. *Trinidad terrain and its main mountainous ranges.*

the northern area of the Nariva Swamp, a freshwater marshland where many seasonal streams from the Central Range occur. Geologically, the soil is mainly composed of the greyish-brown sedimentary sandstone of the Brasso-Manzanilla Formation (Nieweg and Dorst 2001).

The site was first mentioned in the 1940s (Bullbrook 1941, cited by Boomert 2000), with T. Cambridge conducting the first excavations in 1963. He described artifacts that could link the Amerindian settlement to the Late Cedrosan Saladoid cultural complex with a Barrancoid component. A year later, a priest named Rodriguez excavated several test pits that yielded ceramics associated with the Palo Seco culture. This chronocultural background has since been confirmed during surface collecting by P. O'Brien (Nieweg and Dorst 2001).

Within this context, fieldwork began in 1997 (Leiden University) marking out the

limits of the shell midden (Nieweg and Dorst 2001). The team excavated three test pits from different areas of the hill in order to establish the stratigraphy and chronology of the settlement. The archaeological material (ceramics, lithics, and worked shell) indicated two distinct occupations. The first phase, and most abundant in each test pit, presented the characteristics of the Late Cedrosan Saladoid culture (Palo Seco series; about 350–650 AD). The second phase was linked to the Arauquinoid culture (Bontour series; about 650–1400 AD) (Figure 5), with evidence being found exclusively in the upper levels of tests pits 1 and 3. Radiocarbon dates from charcoal samples confirmed this chronology (Table 1).

Later (between 2001 and 2007), an extensive excavation was carried out (500 m²) in which several archaeological features were identified, including postholes and pits. During the Palo Seco phase, two distinct mid-

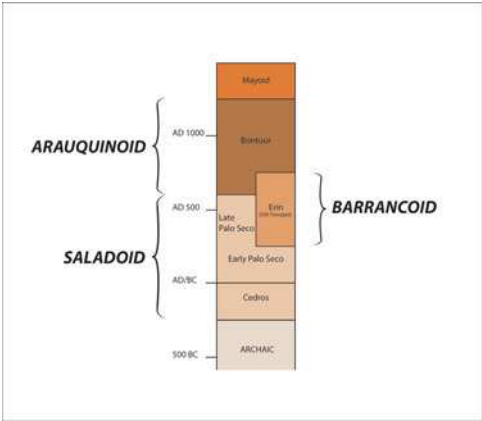


Figure 5. Main cultural phases in Trinidad (Boomert 2000).

dens were identified, suggesting the presence of at least two houses located on the eastern part of the area. Five burials could be linked to this earlier phase. A series of postholes and pits from two buildings characterized the Bontour settlement. The central posthole of one of these houses contained a secondary burial. In the immediate vicinity of these two features were found several burials that could be associated with the Arauquinoid phase. As mentioned by Altena (2007), this funeral pattern suggests a family-based burial orientation. Recent investigations using isotopic data from human remains, gave some information about the foodways of the pre-Columbian settlers of Manzanilla (SAN-1), emphasizing a diversified diet (Healy et al. 2013). Radiocarbon dating of these bones confirmed the long time

span of the occupation, from Late Cedrosan Saladoïd to Arauquinoid cultures, although the continuity of the settlement could not be established.

Trinidad Faunal Framework and Biozones. As previously mentioned, Trinidad is biogeographically part of the South American continent. Many species that can be found on the mainland are indeed present, unlike the other islands of the West Indies, and its flora and fauna are highly diversified. More specifically, the island species spectrum shares many similarities with the nearby coastal region of Venezuela (Boomert 2000; Wing 1962). Such ecological patterns are also present on its sister island Tobago, despite a poorer spectrum of vegetal and faunal species. Recent zooarchaeological data from the island indicates that during Pre-ceramic and Ceramic periods, people relied on both marine resources and a wide range of terrestrial species, including medium-sized and small-sized mammals (*Pecari tajacu*, *Cuniculus paca* now extinct on Tobago) (Steadman and Jones 2006; Steadman and Stokes 2002).

On Trinidad, the Lissamphibians constitute the smallest group of vertebrates with around 30 species from nine families of anuran (Kenny 2008; Schwartz and Henderson 1991).

The reptiles include approximately 85 species (Kenny 2008; Schwartz and Henderson 1991). Within the Squamates there are six families of snakes, of which approximately 40 species can be found in Trinidad. There are five families of lizards, of which around 25 species inhabit Trinidad and Tobago. The rest of the reptilian fauna includes

Table 1. ¹⁴C samples for Manzanilla site.

No.	Lab. no.	Feature	Material	Age BP	Calibrated date *	Source
1	GrA-13865	1-A-14	Charcoal	1590 ± 40BP	cal AD 406-556	Nieweg and Dorst 2001
2	GrA-13866	1-B-7	Charcoal	39,000 ± 500 BP	(rejected)	Nieweg and Dorst 2001
3	GrA-13867	1-B-4	Charcoal	1220 ± 40BP	cal AD 688-892	Nieweg and Dorst 2001
4	Beta-193442	Ft. 16	Human bone	630 ± 40 BP	cal AD 1290-1410	Healy et al. 2013
5	Beta- 193443	Ft. 18	Human bone	620 ± 40 BP	cal AD 1290-1410	Healy et al. 2013

*all 2 sigma, 95% probability

members of the order Testudines: five types of sea turtle, seven terrapins, and two tortoises. A single crocodilian species should also be mentioned: *Caiman sclerops*.

Approximately 400 species constitute the bird group, of which around 250 breed locally: 66 families in 20 orders are represented in Trinidad and Tobago (Ffrench 1991; Kenny 2008).

Finally, a total of 95 species of mammals, including marine mammals can be found on Trinidad: nine orders and about 27 families are represented. All can be considered typical of the adjacent mainland and the wider Neotropics (Eisenberg 1989; Kenny 2008). Bats (60 species) and small rodents (17 species) predominate (Goodwin and Greenhall 1961). The rest of the mammalian group includes six marsupials (Didelphidae), two anteaters (*Cyclopes didactylus*, *Tamandua tetradactyla*), one armadillo species (*Dasybus novemcinctus*), two primates (*Alouatta*, *Cebus*), five carnivores, two artiodactyls (*Mazama americana*, *Pecari tajacu*), and one manatee (*Trichechus manatus*).

Arboreal mammals such as the prehensile-tailed porcupine (*Coendou prehensilis*), the three-toed anteater (*Tamandua longicaudata*) and two types of monkey (*Alouatta seniculus*, *Cebus albifrons*) dwell in the upper levels of the forest. On the ground, the biggest mammals are the red brocket (*Mazama americana*) and the collared peccary (*Pecari tajacu*). Tapirs (*Tapirus terrestris*) are known to have been present in the Trinidad rain forests up to the nineteenth century, when they became extinct (Wing 1962). Caviomorphs such as paca (*Cuniculus paca*) and agouti (*Dasyprocta agouti*) are also commonly found. Opossums (*Didelphis marsupialis*, *Caluromys philander*), the nine-banded armadillo (*Dasybus novemcinctus*), and many types of rice rats and snakes live in the under storey of these woods. The diversity of the fauna decreases in accordance with the density of the forest. Terrestrial reptiles such as iguanas (*Iguana iguana*), snakes and lizards are typical of more open environments like the savanna and the edge of forests. Many authors mention the late

presence of white-tailed deer (*Odocoileus virginianus*) in such milieu without being able to fully determine when it disappeared from the island (Bacon 1978; Boomert 2000; Wing 1962).

Swamps and savanna grasslands occupy the lowland landscapes. Marshes are particularly rich in faunal biomass. Mammals dwelling in this environment include the crab-eating raccoon (*Procyon cancrivorus*), otters (*Lutra enudris*), and water rats (*Nectomys squamipes*). Formerly, the capybaras (*Hydrochaeris hydrochaeris*) also lived on Trinidad (Boomert 2000). Manatees (*Trichechus manatus*) once lived in areas near freshwater swamps, such as Nariva, but they are now almost extinct (Bacon 1978). Alongside birds like the muscovy duck (*Cairina moschata*), which occasionally nests in these regions, many freshwater reptiles can also be found: spectacled caiman (*Caiman sclerops*), terrapins (*Geomyda punctularia*, *Kinosternon* sp.), and snakes like the boa (*Boa constrictor*). Catfish (Ariidae) and snooks (Centropomidae) are also relatively abundant in these brackish waters. And finally, crabs live on the shores of the freshwater swamps and rivers.

Although there have been no recent taxonomic reviews of the marine fish of Trinidad and Tobago, it is possible to evaluate the ichthyofauna richness using the 950 species in several dozens of orders and families (Kenny 2008; Wing and Reitz 1982).

General Setting of the Manzanilla Faunal Assemblage

Taphonomy and Bone Preservation. The taphonomical study of this assemblage included analysis of the fragmentation and the inventory of attritional and anthropogenic marks. Fragmentation may be a major bias in specimen quantification (Lyman 1994 2008:205). It can have various causes ranging from pre-depositional to post-depositional processes (e.g., butchering practices, trampling, diagenetic agents). Osseous fragmentation is a crucial issue since it may affect specimen identification and taxonomic quantification through the increase of the NISP (Cannon 2013). Various meth-

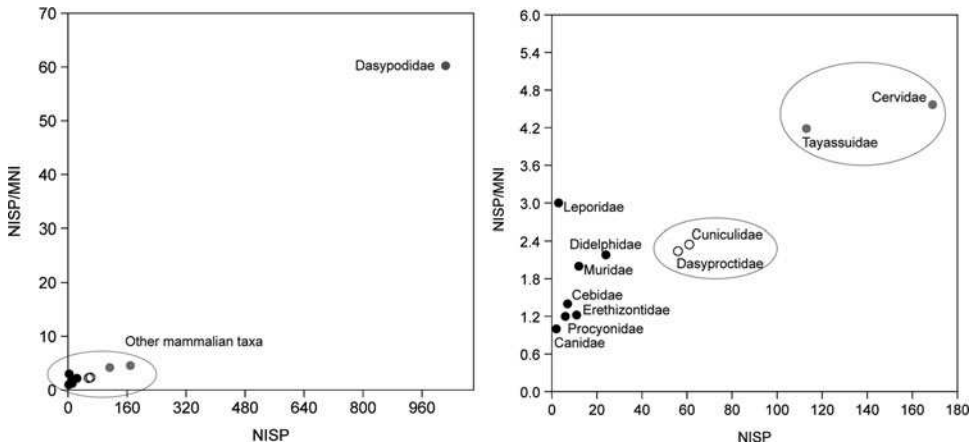


Figure 6. Fragmentation indices of Manzanilla assemblage (left: with *Dasypodidae*, right: without *Dasypodidae*).

ods can be employed to evaluate the degree of fragmentation within an assemblage; the Klein and Cruz-Urbe (1984) NISP/MNI ratio is commonly used to provide such an estimate. The main idea behind this method is that NISP increases with higher fragmentation, while MNI should stay stable. It should be noted, however, that some biases exist with this method, as outlined by Lyman (2008) and more recently by Cannon (2013). Indeed, with higher fragmentation the number of fragments increases up to the point that some specimens are no longer identifiable. The solution proposed by Cannon to infer fragmentation with more accuracy is to measure specimen size. Unfortunately, we were unable to apply this new approach on the Manzanilla assemblage, though it appears to provide interesting results.

The NISP/MNI ratio was calculated for mammalian taxa, which were the most numerous terrestrial vertebrates in the assemblage (Figure 6). The first observation concerns *Dasypodidae*, which unsurprisingly presented an extremely high degree of fragmentation due to the numerous carapace specimens present in the assemblage ($n = 970$). Some caviomorphs, like pacas and agoutis, cervids, and peccaries appeared with an intermediate rate of fragmentation. The small-sized mammal subset is the least fragmented category.

The set itself (Table 2) is well preserved with little evidence of weathering, as none of the bones presented stage 2 state (Behrens-meyer 1978). The percentage of bone modification marks is also low with the only marks being noticed relating to pre-depositional processes like rodent gnawing traces, burning, and cutmarks.

Teeth traces corresponding to small-sized rodents, like rice rats, mean some bones were within reach of small animals dwelling near the human settlement. This evidence could suggest the existence of commensalism among small rodents and porcupines in pre-Columbian Manzanilla.

Evidence of burning (linked to cooking habits and/or waste management), was also scarce within the osseous material of Manzanilla, and detectable in only 10 specimens (0.55 % of the total NISP). Burn marks were generally related to big-sized and medium-sized mammals: *Mazama americana* (NISP = 4), *Tayassu* sp. (NISP = 2) and *Dasyprocta agouti* (NISP = 2). In eight cases the burnt bones belonged to animal limbs, often epiphyseal parts. This fact reflects potential cooking practices: roasting meat pieces directly in the fire or embers. Other anthropogenic cutmarks were also scarce, and mainly affected Cervidae and Tayassuidae specimens. In most cases, these marks can be linked to cutting or carving activities.

Table 2. Some taphonomic variables of Manzanilla faunal assemblage.

Taphonomic variables	Indices
NISP/MNI	8.68
Big game/small game ratio	0.16
Percent weathered (> stage 2)	0
Percent abraded	0
Percent root etching	0
Percent carnivore marks	0
Percent rodent marks	0.33
Percent burned	0.55
Percent cut marks	0.49

There were some incisions on the base of *Mazama* antlers, however, possibly linked to an attempt to separate the bone from the skull.

The generally high preservation of the specimens is most likely due to the abundance of shellfish remains which improved preservation conditions (alkaline pH) due to the precipitation of carbonate particles in the sediment (Reitz and Wing 2008:141).

Marine Resources

The total NISP is 3,920, representing 40 different taxa (Table 3). The faunal spectrum is defined by the presence of five classes of vertebrates including Actinopterygii, Chondrichthyes, Mammalia, Reptilia, Aves, and Amphibia. Numerically, fish remains were more abundant than any other taxa.

As previously mentioned, we chose to focus on tetrapods in order to shed some light on pre-Columbian hunting practices in Manzanilla, but it must be noted that marine resources also make up an important part of the faunal assemblage. Indeed, with a total NISP of 2,104, fish remains constitute almost 54% of the total assemblage; other marine resources, such as mollusks, were also numerous. Most of them came from brackish waters and mud sandy bottoms: Ariidae (*Arius proops*), Albulidae (*Albula vulpes*), Elopidae (*Elops saurus*), Centropomidae (*Centropomus* sp.), Malacanthi-

dae (*Malacanthus plumieri*), Haemulidae (*Conodon nobilis*), and Sciaenidae; reefs and rocky bottoms: Muraenidae, Holocentridae, Serranidae, Lutjanidae, Haemulidae (*Haemulon* sp.), Sphyraenidae (*Sphyraena* sp.), Scaridae (*Sparisoma* sp.), and Diodontidae; or pelagic surface waters: Carangidae (*Caranx* sp.) and Scombridae. Analyzing the diet of the Manzanilla former inhabitants would require the studying of each faunal remain (including invertebrate and especially shellfish), which, according to Healy et al. (2013), represent 90% of the faunal material recovered from the deposit. Several species compose the invertebrate spectrum (Nieweg and Dorst 2001), but the coquina clam (*Donax striatus*) dominates. The trigonal tivala (*Tivela mactroides*), Brazilian ark (*Anadara brasiliiana*), West Indian crown conch (*Melongena melongena*), green-star shell (*Lithopoma tuber*), and queen conch (*Lobatus Strombus costatus*) were also discovered. Finally, some remains of freshwater oysters (*Crassostrea rhizophorae*) were noted.

RESULTS

Faunal Spectrum of Tetrapods

The tetrapods were identified on the basis of 1,807 NISP osseous remains, which represents approximately 46% of the whole assemblage (Table 4, Figure 7). They are dis-

Table 3. Quantification in NISP of the whole assemblage.

		NISP	% NISP
Chondrichthyes		9	0.23%
Rajiformes	Dasyatidae	9	0.23%
Actinopterygii		2104	53.67%
Elopiformes	Elopidae	6	0.15%
Albuliformes	Albulidae	77	1.97%
Anguiliformes	Muraenidae	4	0.10%
Siluriformes	Ariidae	491	12.54%
Beryciformes	Belonidae	14	0.36%
Beryciformes	Holocentridae	40	1.02%
Perciformes	Centropomidae	33	0.84%
Perciformes	Serranidae	7	0.18%
Perciformes	Malacanthidae	2	0.05%
Perciformes	Carangidae	147	3.75%
Perciformes	Lutjanidae	15	0.38%
Perciformes	Haemulidae	68	1.74%
Perciformes	Sciaenidae	16	0.41%
Perciformes	Scaridae	8	0.20%
Perciformes	Sphyraenidae	3	0.08%
Perciformes	Scombridae	1	0.03%
Tetraodontiformes	Diodontidae	23	0.59%
	ND	1149	29.33%
Amphibia		6	0.15 %
Anura	Bufonidae	6	0.15%
Reptilia		248	6.33 %
Testudines	Testudinidae	72	1.84%
Testudines	Cheloniidae	52	1.33%
Crocodylia	Alligatoridae	42	1.07%
Squamata	Iguanidae	68	1.74%
Squamata	Boidae	3	0.08%
Squamata	Colubridae	2	0.05%
Squamata	Viperidae	5	0.13%
Squamata	Serpentes	2	0.05%
	ND	2	0.05%
Aves		59	1.51 %
Galliforme	Cracidae	2	0.05%
Gruiforme	Rallidae	1	0.03%
	ND	56	1.43%

(Continued on next page)

Table 3. Quantification in NISP of the whole assemblage. (Continued)

		NISP	% NISP
Mammalia		1494	38,11 %
Didelphimorphia	Didelphidae	24	0.61%
Xenarthra	Dasypodidae	1024	26.12%
Primates	Cebidae	11	0.28%
Carnivora	Canidae	2	0.05%
Carnivora	Procyonidae	6	0.15%
Artiodactyla	Tayassuidae	113	2.88%
Artiodactyla	Cervidae	169	4.31%
Rodentia	Muridae	12	0.31%
Rodentia	Erethizontidae	7	0.18%
Rodentia	Cuniculidae	61	1.56%
Rodentia	Dasyproctidae	56	1.43%
Rodentia	Dasyproctidae/Cuniculidae	6	0.15%
Rodentia	Leporidae	3	0.08%
Total		3920	100%

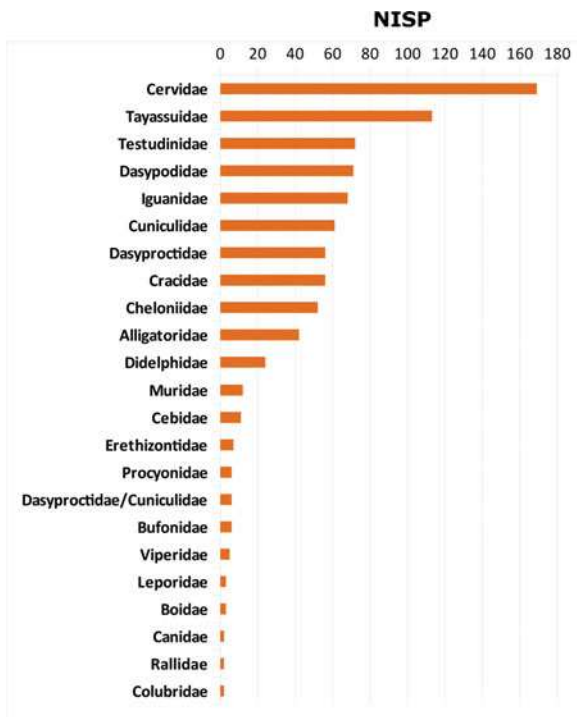


Figure 7. Bar plot of Manzanilla tetrapod NISP's (NISP Dasypus = 71, corrected according MNI_Dasypus = 17).

Table 4. Quantification in MNI of the tetrapod taxa identified in Manzanilla.

		NISP	%NISP	MNI	%MNI	BM (g)	%BM
Amphibia							
Anura	Bufonidae	6	0.28%	2	0.86%	1.2	0.02%
Reptilia							
Testudines	Testudinidae	72	4.02%	11	4.72%	63.1	2.47%
Testudines	Cheloniidae	52	2.91%	11	4.72%	213.3	8.35%
Crocodylia	Alligatoridae	42	2.35%	10	4.29%	88	3.44%
Squamata	Iguanidae	68	2.91%	15	5.15%	39	1.45%
Squamata	Boidae	3	0.17%	3	1.29%	1.16	0.05%
Squamata	Colubridae	2	0.11%	2	0.86%	0.2	0.01%
Squamata	Viperidae	5	0.28%	2	0.86%	0.4	0.02%
Squamata	Serpentes	2	0.06%		0.00%	0.2	0.00%
	ND	2	0.11%		0.00%	0.2	0.01%
Aves							
Galliforme	Cracidae	2	0.11%	1	0.43%	2.05	0.08%
Gruiforme	Rallidae	1	0.06%	1	0.86%	0.6	0.02%
	ND	56	3.13%	5	2.15%	9.9	0.39%
Mammalia							
Didelphimorphia	Didelphidae	24	1.34%	11	4.72%	22.5	0.88%
Xenarthra	Dasypodidae	1024	57.24%	17	7.30%	156.05	6.11%
Primates	Cebidae	11	0.61%	9	3.86%	19.1	0.75%
Carnivora	Canidae	2	0.11%	2	0.86%	14.9	0.58%
Carnivora	Procyonidae	6	0.34%	5	2.15%	6.9	0.27%
Artiodactyla	Tayassuidae	113	6.32%	27	11.59%	410.5	16.06%
Artiodactyla	Cervidae	169	9.45%	37	15.88%	1278.3	50.02%
Rodentia	Muridae	12	0.67%	6	2.58%	1.6	0.06%
Rodentia	Erethizontidae	7	0.39%	5	2.15%	15.5	0.61%
Rodentia	Cuniculidae	61	3.41%	26	11.16%	143.8	5.63%
Rodentia	Dasyproctidae	56	3.13%	25	10.73%	64.2	2.51%
Rodentia	Caviomorpha	6	0.34%	1	0.43%	1.3	0.05%
Rodentia	Leporidae	3	0.17%	1	0.43%	4.3	0.17%
Total		1807	100.00%	235	100.00%	2558.26	100%

tributed among Mammalia (NISP = 1,494, MNI = 172), Reptilia (NISP = 248, MNI = 54), Aves (NISP = 59, MNI = 8), and Lissamphibia (NISP = 6, MNI = 2). A total amount of 25 taxa were eventually identified from this set.

Mammals. As shown in Table 2, mam-mal remains clearly dominate the collection. Fifteen taxa have been identified from this sub-sample according to their relative impor-tance using MNI counts (Figure 8):

- Cervidae: only one species of Cervi-dae was recorded from the assemblage: *Mazama americana* (brocket deer), represented by at least 37 individu-als. These small-sized artiodactyls usu-ally dwell in thick areas of tropical rain forests. No specimens of white-tailed deer (*Odocoileus virginianus*), now ex-tinct on the island (Bacon 1978; Wing 1962), were identified in the faunal set.

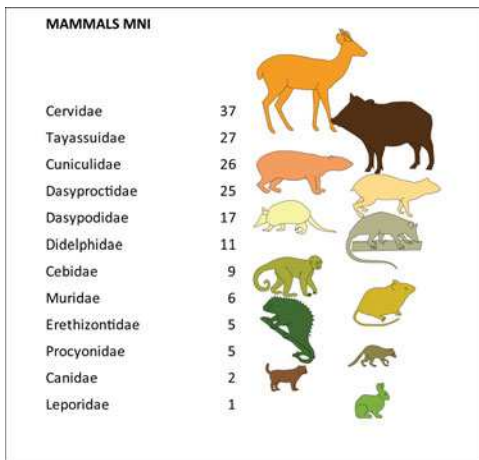


Figure 8. Table of MNI counts of mammal bones.

- **Tayassuidae:** at least 27 individuals were identified. The specific MNI count regarding peccaries shows the prevalence of *Pecari tajacu* (MNI = 8) over *Tayassu pecari* (MNI = 1). This is probably due to the fact that the white-lipped peccary usually lives in large packs which travel across a wide home range. Furthermore, its behavior is fiercer than the collared peccary, which makes hunting it more hazardous. In contrast, collared peccaries live in smaller territories and can be easily found feeding on cultivated crops (Emmons and Feer 1999; Pérez and Pacheco 2006).
- **Caviomorpha:** this sub-order includes two species of large-sized rodents: the paca (*Cuniculus paca*) and the agouti (*Dasyprocta agouti*). Although quite similar in shape, these two species have distinct behavioral habits. Usually nocturnal, the paca is most commonly found near water and swampy areas, but can also be seen in open forest far from water. Agoutis are more active during the day and usually dwell in open forests (Emmons and Feer 1999). They were relatively abundant in the Manzanilla assemblage with a total of 123 specimens associated to caviomorphs, equally dis-

tributed between both species. Finally, another caviomorph worth noting is the prehensile-tailed porcupine (*Coendou prehensilis*, MNI = 5).

- **Oryzomyini:** scant remains (MNI = 6) of rice rats attest to the presence of small rodents in Manzanilla.
- **Dasypodidae:** *Dasybus novemcinctus* were also very common in the deposit. The abundance of armadillo remains (NISP = 1,024) was evidenced by the presence of high numbers of dorsal plate bones. Indeed, this thin skeletal part of the armadillo dramatically increased the amount of remains belonging to this species, but the MNI helped put this count into perspective (MNI = 17). This species is chiefly nocturnal and is commonly found throughout forests and thickets.
- **Primates:** at least 9 individuals belonging to the Cebidae family were identified. Both red howler monkeys (*Alouatta seniculus*, MNI = 7) and capuchin monkeys (*Cebus albifrons*, MNI = 2) which live in the upper levels of the tropical rain forest and are preferably diurnal.
- **Carnivores:** crab-eating raccoons (*Procyon cancrivorus*, MNI = 5) and bush dogs (*Speothos venaticus*, MNI = 2) complete the mammal remains collection.

A Rare Species in Archaeological Deposits. Two bones, one upper M3 found in Unit-2 (context 2D4) and a right jawbone found in Unit-1 (context 1A4), indicated the presence of *Speothos venaticus* (Figure 9).

Speothos venaticus, or bush dog, is a small-sized canid (4 to 7 kg) with an elongated body and short legs (de Mello Beisiegel and Zuercher 2005). It is widely distributed across northern South America, but its population is highly scattered and occurrence data is still extremely rare. They have inter-digital membranes that are adapted for the soft soil of riverside environments. Bush dogs are carnivorous and dwell in forest edges or near freshwater streams.

References to *Speothos venaticus* in the archaeological record are extremely scarce and, as far as we know, no specimens of

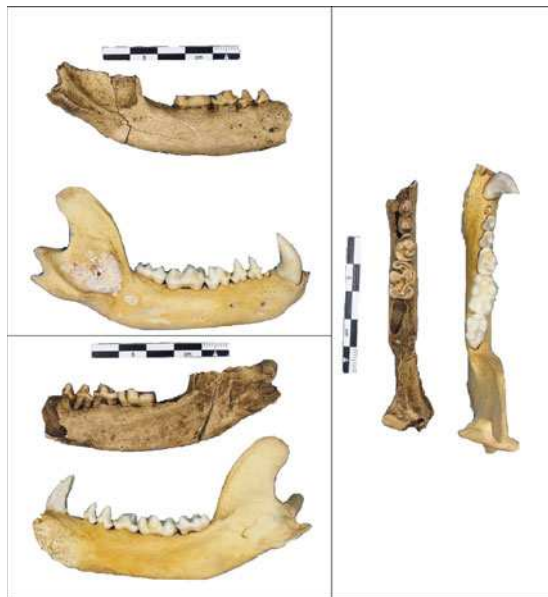


Figure 9. *Speothos venaticus* right jawbone from Manzanilla-1 site compared with specimen from MNHN collections (973-96M-0002, French Guyana).

this species are mentioned from archaeological contexts in the region. According to IUCN red list data, there is no evidence for bush dogs on the island of Trinidad (<http://maps.iucnredlist.org/map.html?id=20468>). It is, however, known to dwell in nearby Venezuela, especially in southern parts of the country, though it also appears to be present in the north (Strahl et al. 1992). Therefore, the type of relationship maintained with human populations, and particularly with pre-Columbian Amerindians, is almost unknown. Some authors (Herskovitz 1969; Stahl 2012) postulate the bush dog could have been introduced by humans in some parts of its current distribution range, such as Panama; however, these assertions are based on biogeographical facts rather than on archaeological records.

Birds, Reptiles and Lissamphibians. *Aves* are quite scarce in the collection, and due to the high degree of bone fragmentation we were unable to determine them with great accuracy: members of the Cracidae family (NISP = 2) and at least one individual of the Rallidae (rails) family (NISP = 8).

With 248 identified specimens and 54 individuals, the Reptilia are the second class of tetrapod represented in the deposit. Green iguana (Iguanidae, *Iguana iguana*) is the most common Squamata found on the Manzanilla site with at least 15 individuals. The remains of 114 were associated with land (Testudinidae, MNI = 11) and sea turtles (Cheloniidae, MNI = 11). Sea turtle vestiges came entirely from the *Chelonia mydas* species. These taxa are currently very common in the Manzanilla area: the beaches of Cocos Bay, close to the settlement, are known to shelter many green turtle nesting sites (Bacon 1978; Boomert 2000). The remains of 42 bones belonging to 10 individuals of the spectacled caiman were also identified (*Caiman sclerops*). It is usually found in riverine environments though it also tolerates salt water. Some snake bones (*Boidae* and *Colubridae*) were also found, though these were scarce (NISP = 12). Amphibians are also rare in the assemblage and were represented by only one species: the cane toad (*Rhinella (Bufo) marinus*, MNI = 2).

Chronological Variations. The compilation of the faunal remains, according to the

I, II, and III stratigraphic contexts, provides information about the taxa distribution evolution.

There is a strong increase of faunal remains from Set II, the median part of the stratigraphy (Figure 10), concerning mainly fish and mammal remains. Figure 10 illustrates that dependence on fish seems to have increased with time, while mammal remains stayed stable. As emphasized before, given the potential biases, caution should be taken regarding any further conclusions with these counts.

The MNI count (Figure 11) also shows a strong increase of faunal remains during the second stratigraphic phase, most notably for Cervidae, Dasyproctidae, and Iguanidae. In contrast, the number of peccaries seems to remain constant. Finally, the latter stratigraphic contexts appear to be marked by a strong rise in the MNI of the Cheloniidae, corroborating initial impressions that Manzanilla's last inhabitants relied primarily on marine resources. Overall, the composition of the Manzanilla assemblage shows an increase in marine and semi-aquatic resources during the time-span of the human occupation (Figure 12).

The results show the relative richness of the assemblage (dI) decreases slightly over time. In contrast, the specific diversity index

(H') records an important loss of diversity in the last phase of the settlement. This could be linked to a greater reliance on marine resources, and therefore a narrower range of hunted species.

Hunting and Ecological Niches

Recent isotope analysis on human remains found in Manzanilla show that the Amerindian diet relied heavily on marine resources (shellfish, fish) and plants like manioc or maize (Boomert 2000; Healy et al. 2013).

However, as mentioned previously, an initial look at the faunal assemblage shows that a significant part of the identified remains can also be associated to terrestrial animals (43.6% of the NISP). The wide diversity of ecological niches exploited by the pre-Columbian population (Figure 13) suggests a lifestyle based on an opportunistic exploitation of the immediate surroundings, involving fishing, hunting, and collecting. The practice of horticulture should also be considered, given the results of isotopic studies and the current archaeological record (Boomert 2000:95–100).

When calculating edible meat masses using allometric formulae (Reitz and Wing

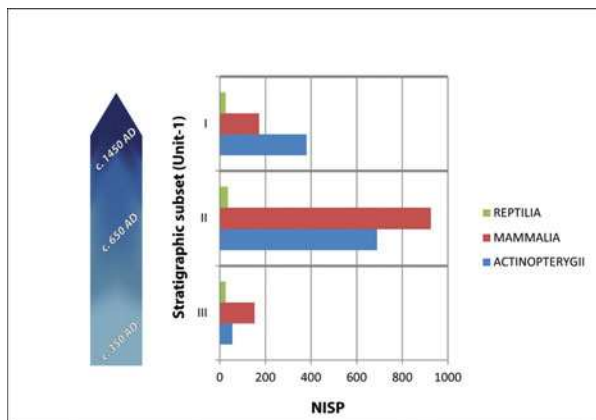


Figure 10. Evolution between stratigraphic set of the faunal spectrum composition in test pit No. 1 (aves, lissamphibians, and chondrichthyes NISPs are too low to be significant on this graph).

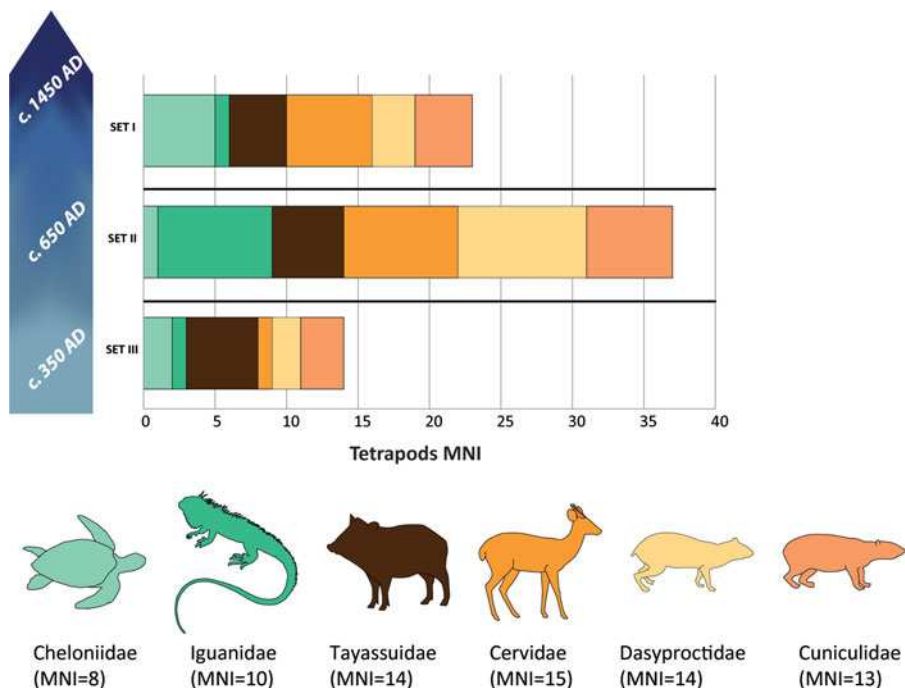


Figure 11. Evolution of some tetrapod's MNI depending on stratigraphic set in Unit-1.

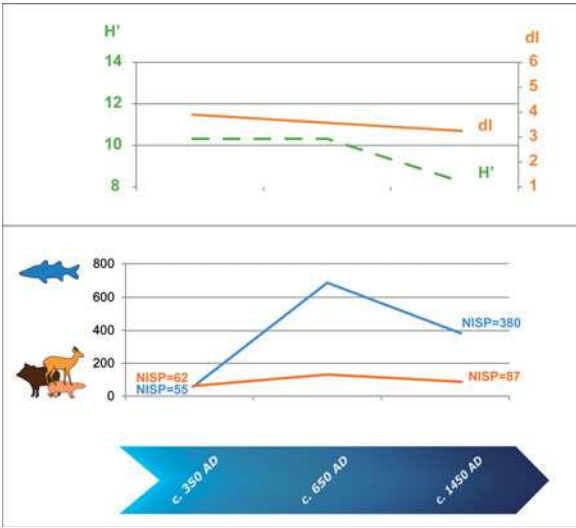


Figure 12. Synthesis of faunal assemblage evolution in Manzanilla. Upper graph shows changes in species diversity (H' , Simpson index converse) and richness (dl , Margalef index). Below, the graph reveals the progression of NISP count between fishes and mammals.

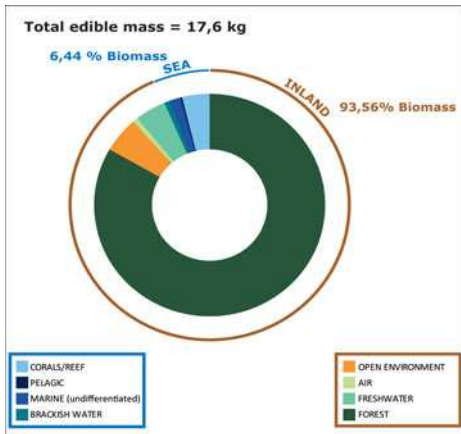


Figure 13. Ecological niches represented in the Manzanilla faunal sample (% of edible biomass).

2008:236–241), the proportion of marine vertebrates dramatically drops. The origin of terrestrial animals eaten by the former inhabitants of Manzanilla gives valuable information on hunting habits. By crossing the MNI counts with environmental data, regarding each taxa (habitat, locomotion), we can see that hunters exploited a wide spectrum of environments (Figure 14). More specifically, hunting activities concentrated upon species living in lower levels of tropical rain forest, such as brocket deer, and medium-sized mammals dwelling both in the forest environment and more opened areas (agoutis, pacas, peccaries).

Cervids, Tayassuids, and Caviomorphs Skeletal Parts

The percentage represented by greater anatomical parts (head, axial skeleton, forelimb, hindlimb, hand/feet bones) provides evidence on carcass rendering strategies across the four main terrestrial taxa identified on Manzanilla (brocket deer, peccary, agouti, and paca). The results highlight a tenuous difference between the two species of caviomorphs (Figure 15), and seem to indicate an equal treatment of both agouti and paca carcasses in butchering and refuse dis-



Figure 14. Ancient environment around Manzanilla site (Beard 1946).

carding practices. Upper limb parts are well represented as are skull bones. The quasi-absence of axial and hand/foot bones suggests that the caviomorph carcasses were butchered elsewhere, and that this area represents the remains of a cooking refuse deposit.

Results regarding the comparison of bigger sized mammals, cervids and peccaries, show a similar representation pattern with some slight differences (Figure 16). Upper limb bones, which represent a higher meat yield (Binford 1978; Metcalfe and Jones 1980), are also abundant. The case of a single skull part presents a slight discontinuity, especially for tayassuids. As for caviomorphs, hand/foot bones are quite scarce. Evidence of spinal parts, especially for cervids, is interesting as although not as numerous as limb bones, axial bones of *Mazama americana* exist in significant numbers in the collection. This could indicate the rendering of brocket deer carcasses in the area, with the remnants being discarded close to the settlement. Faith and Gordon's method on game rendering and transport complements our view on artiodactyls carcass management (Figure 17, Tables 5 and 6) (Faith and Gordon 2007).

Despite Spearman's rho values being quite low, intimating a lack of significant results mainly due to sample size, peccaries

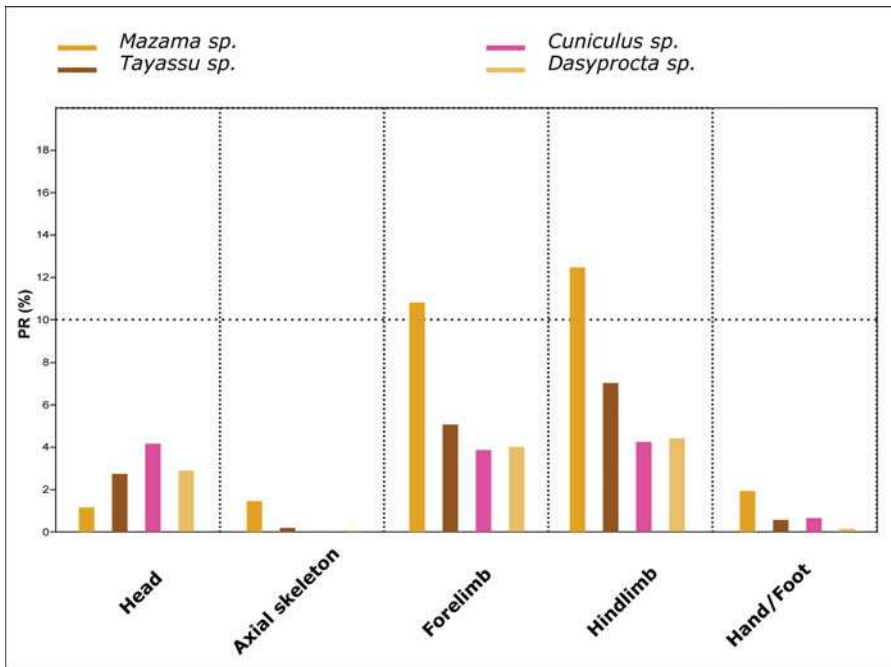


Figure 15. Comparison of the distribution of skeleton elements between artiodactyls and caviomorphs.

and deer subsets do show some interesting trends. The frequency of high survival elements in comparison with the Standardized Food Utility Index (SFUI) shows the evenness indexes falling in the range of “unbiased” transport strategy. In other words, it is likely that the anatomical parts of cervids and tayassuids would have been transported in direct proportion to their economic utility.

Artiodactyls Osteometry

Osteometrical studies are relatively scarce in Caribbean and Neotropical zooarchaeology, as is the data to make comparisons between archaeological deposits and modern faunas. In the Manzanilla case, we believe such an approach is relevant to bring out the potential influence of insularity for big and medium-sized mammals. The study of insular dwarfism has been approached in various paleontological and archaeological studies in other parts of the world. On the

Mediterranean island of Cyprus, an island never connected to the mainland, the study of the paleontological deposit of Aetokremnos showed that a population of imported suids, much smaller than their mainland counterparts, inhabited the island during the twelfth millennium BC (Vigne et al. 2009). Again in Cyprus, the Prepottery Neolithic site of Shillourokambos yielded important faunal remains (Vigne 2011). Metric analyses on ungulate populations, especially cervids (*Dama dama mesopotamica*) and suids (*Sus scrofa*), confirm this trend of smaller insular individuals. In the Western Mediterranean, early Holocene populations of suids also show a tendency toward dwarfism (Vigne 1988).

Obviously, limited conclusions can be drawn from our small data set (four cervids and four peccaries); nevertheless, the comparisons presented here aim to provide some tracks for further thinking on the evolution of Trinidadian mammalian fauna.

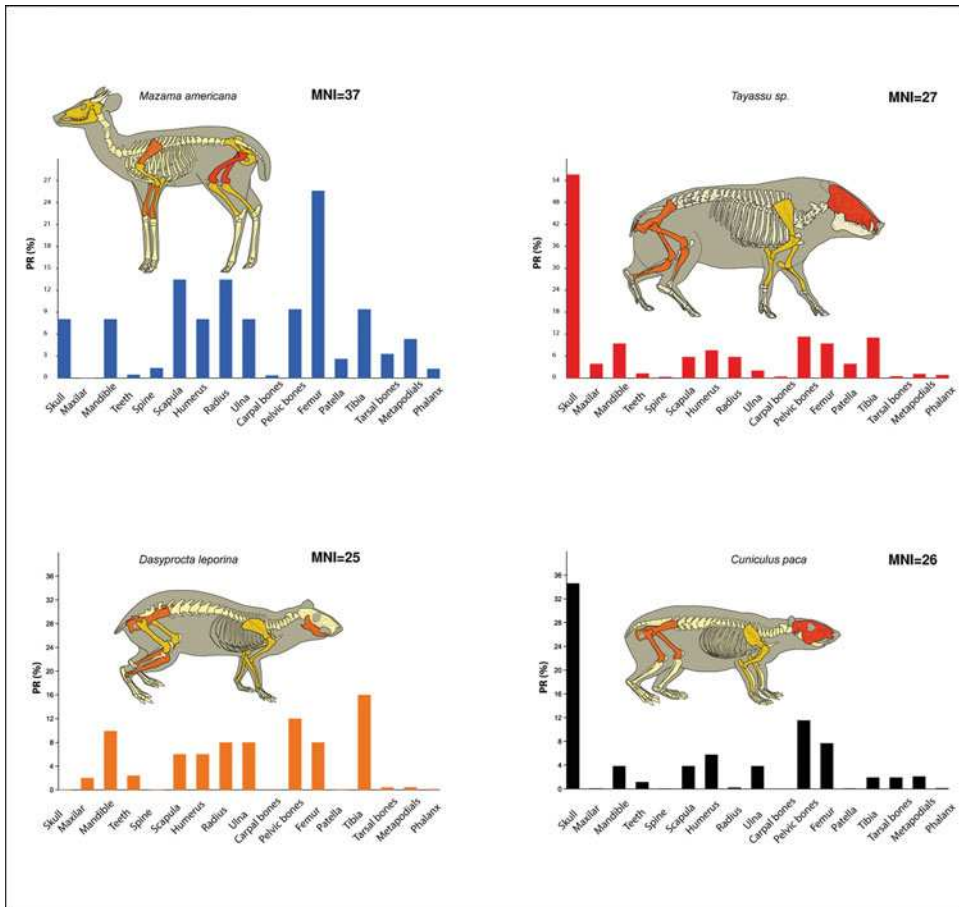


Figure 16. Distribution of artiodactyls and caviomorphs skeletal parts in Manzanilla faunal sample.

For example, the estimation of femur slenderness from a Manzanilla tayassuid tends to indicate that peccaries were somewhat smaller and more slender than the comparison specimens (Figure 18). A similar observation on M3 shows that the *Tayassuidae* found in the deposit tended to be of smaller dimensions. One way to demonstrate this could be by using advanced metrical data to illustrate the allometric relationships between limb bones and their extremities (Vigne 2011). If confirmed, it may indicate the insularity of this population from its continental counterparts at an early stage in its anatomical development.

Osteometrical data for Cervidae was also recorded (Figure 18). Though all cervid bones from Manzanilla stemmed from *Mazama americana*, we still compared them to *Odocoileus virginianus* reference specimens from the MNHN collections. According to Wing (1962), although they are now extinct on Trinidad, white-tailed deer seem to have dwelled on the island during pre-Columbian times (Bacon 1978).

As one can see from the graphs (Figure 18), metrical data is highly scattered. This can be partially explained by the high sexual dimorphism existing among cervids (Post et al. 1999). The small size of the sample

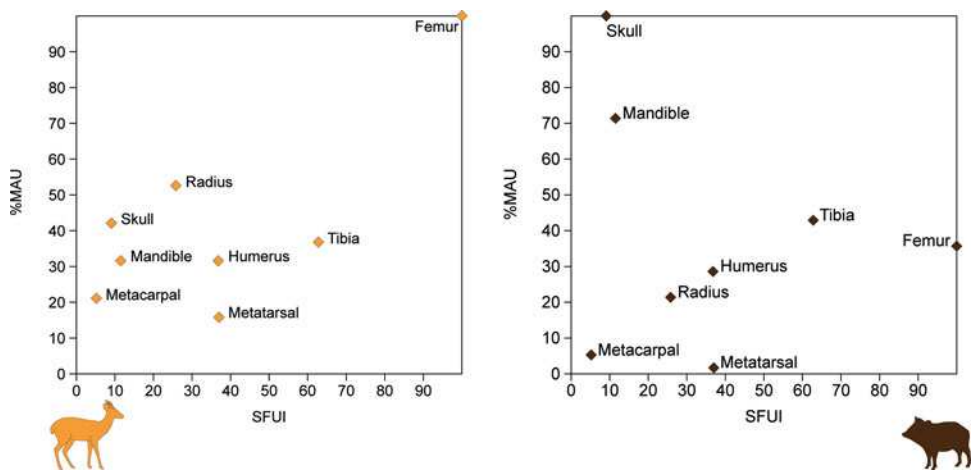


Figure 17. Utility indices (SFUI) crossed with % MAU of artiodactyls of the Manzanilla deposit.

is an obstacle to draw advanced conclusions. However, we can state that red brockets from the Manzanilla deposit appear to be more slender than reference specimens.

DISCUSSION

As previously stated, limited archaeological data regarding faunal remains in Trinidad has been published. The first, and most exhaustive, is from E. Wing’s Ph.D. thesis (1962) which concentrates mainly on mam-

mal remains. Wing’s MNI counts for four pre-Columbian settlements (Cedros, Erin, Palo Seco, and Quinam) show that, as in Manzanilla, the major terrestrial hunted species were brocket deer, collared peccary, agouti, and paca. Nine-banded armadillos and black-eared opossum were also well represented but in lesser proportions.

Due to the diversity of terrestrial fauna in Trinidad, similar only to Tobago within the Caribbean, scientific literature is unable to provide any accurate examples on West Indian hunting practices here (Bacon 1978;

Table 5. Table of skeletal parts counts of cervid remains in Manzanilla.

	SFUI	NME	MAU	%MAU
Femur	100	19	9.5	100.0%
Tibia	62.8	7	3.5	36.8%
Metatarsal	37	3	1.5	15.8%
Humerus	36.8	6	3	31.6%
Radius	25.8	10	5	52.6%
Mandible	11.5	6	3	31.6%
Skull	9.1	4	4	42.1%
Metacarpal	5.2	4	2	21.1%
Shannon evenness (E)	0.843			«Unbiased»

$r_s = 0.31138$ $p = .4528$

Table 6. Table of skeletal parts counts of peccary remains in Manzanilla.

	SFUI	NME	MAU	%MAU
Femur	100	5	2.5	35.7%
Tibia	62.8	6	3	42.9%
Metatarsal	37	1	0.12	1.7%
Humerus	36.8	4	2	28.6%
Radius	25.8	3	1.5	21.4%
Mandible	11.5	5	5	71.4%
Skull	9.1	7	7	100.0%
Metacarpal	5.2	3	0.37	5.3%
Shannon evenness (E)	0.801			«Unbiased»

$r_s = -.07143, p = .86653.$

Pregill et al. 1994). However, zooarchaeological studies in Tobago suggest patterns similar to those observed on Manzanilla (Steadman and Jones 2006; Steadman and Stokes 2002): a high exploitation of marine resources supplemented by terrestrial game in pre-Columbian diets (especially during the preceramic period). Another interesting analogy with Manzanilla is that the preference for big game hunting (peccaries) is not linked to site location (inland or coastal).

Other parallels can be found elsewhere on the Neotropics mainland. In her study of Cerro Brujo (Aguacate Peninsula, Panama), Linares mentions the “garden hunting” pattern (Linares 1976). This food-harvesting strategy consists of hunting animals whose habitat primarily corresponds to forest margins, and that could be attracted towards cultivated crops or gardens; Manzanilla hunting strategies could partly fit this pattern. Recent studies on wild animal crop damage in Bolivia demonstrated that the species most likely to devastate maize or yucca fields were agoutis and peccaries, a taxa well represented on Manzanilla (Pérez and Pacheco 2006; Pérez and Pacheco 2014). On the other hand, the high rate of brocket deer remains in the tetrapod subset implies the exploitation of other ecological niches: *Mazama americana* generally dwell in the thickest parts of the rain forest, far from human activity (Emmons and Feer 1999).

All of these elements seem to suggest that pre-Columbian settlers had a rather opportunistic strategy to harvest animal proteins, a point previously mentioned in the ethnohistorical data (Boomert 2000: 329–334).

Regarding hunting practices, it appears that seasonality had a strong influence on food supplies with a marked discrepancy in animal availability during the dry and wet season, when hunted animals are scarcer. Current ethnographic data show that hunters in the Orinoco basin and the Guyanas mainly use traps to catch game (Rivière 1984:88; Sponsel 1986). They also rely on domesticated dogs to track down big game such as cervids or peccaries (Roth 1924). Unfortunately, no dog remains were found on Manzanilla and records of this species are particularly rare in the Trinidad archaeological record; however, they are mentioned on other West Indian islands (Boomert 2000; Drewett 1991; Grouard 2007). The tools used to hunt these animals include bows, which were used extensively on Trinidad, according to the first European explorers of the region (Cohen 1969; Markham 2010).

Finally, the question of the recent morphological evolution of ungulates on Trinidad has been evoked through the osteometrical study, though the data is still scarce and more measurements should be taken to

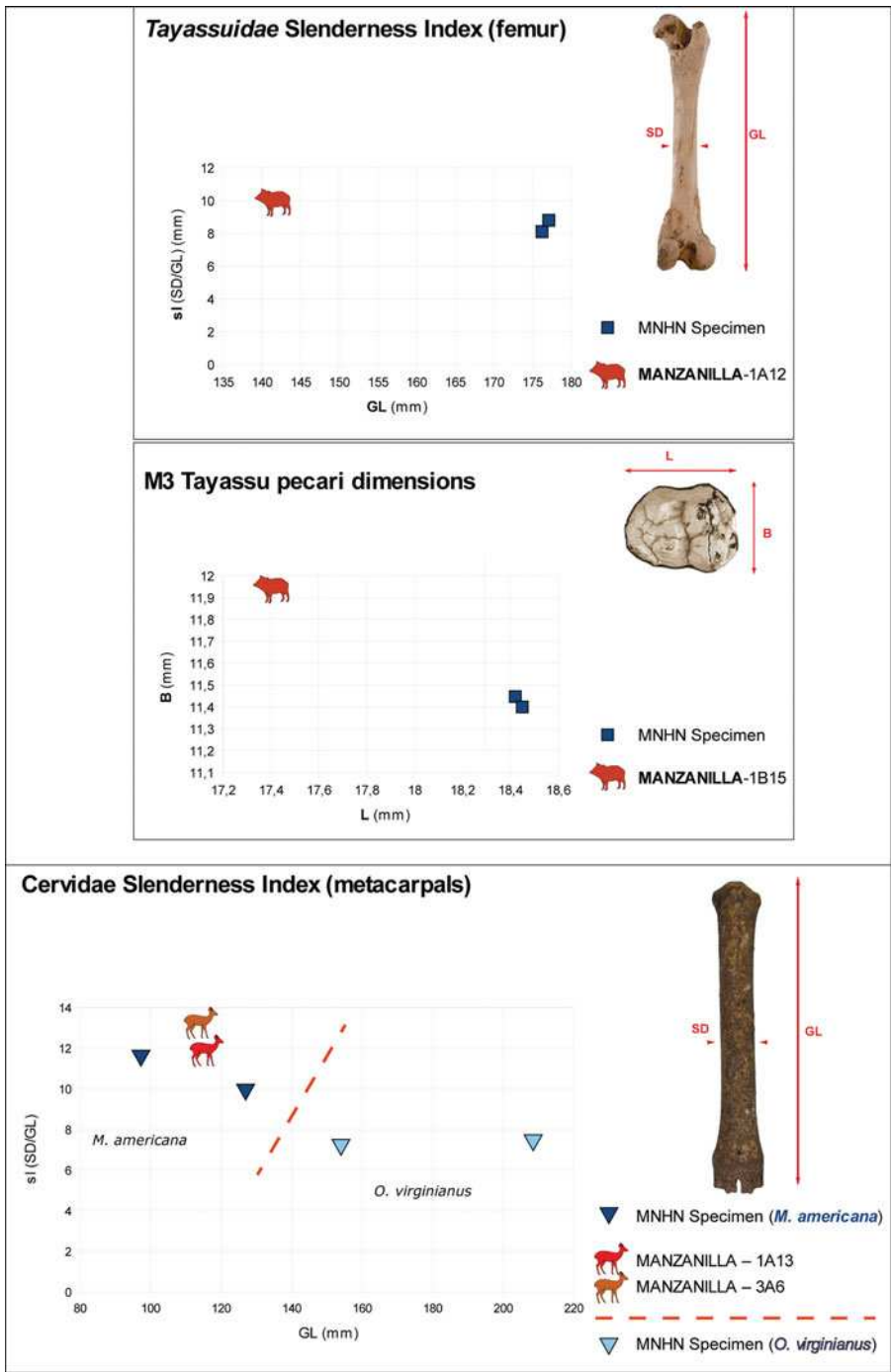


Figure 18. Bone osteometry of the artiodactyls: Comparisons between Manzanilla specimens and MNHN reference collection.

enhance this analysis. Still, an initial trend seems to indicate that Manzanilla artiodactyls were rather small in size. At this point of the study, linking these results with biogeographical phenomena like insular dwarfism is risky but it could be considered as a reflection for further investigations.

CONCLUSION

The Manzanilla hunting practices tie in with the known patterns of terrestrial animal harvesting strategies in the area, connecting Trinidad and Tobago with the mainland rather than with other West Indian islands. The pre-Columbian inhabitants of Manzanilla chiefly hunted mid-sized game like brocket deer, peccaries, and small-sized animals like caviomorphs. The tracked species spectrum shows that hunters exploited a wide range of environments, starting with the immediate surroundings of the site up to the thickest regions of evergreen forest. Finally, the available zooarchaeological data shows that faunal remains correspond to cooking refuse, implying that the butchering of carcasses may have occurred elsewhere. While utility indices tend to outline the fact that the anatomical parts of larger game were transported according to their economic importance.

END NOTES

1. $dI = \frac{(S-1)}{\ln N}$ where S: number of taxa, N: total MNI.
2. $H' = \frac{1}{\sum p_i^2}$ where $p_i = \frac{MNI_{i\text{axon}}}{MNI_{\text{total}}}$.
3. $Y = aX^b$ where Y: estimated sample biomass, X: specimen weight, a: Y-intercept of the linear regression line, b: slope of the regression line.
4. $PR = \left(\frac{FO}{FT \times MNI} \right) \times 100$ where PR: percentage of representation, FO: MNE of each specimen, FT: number of each specimen in a complete skeleton.
5. Ref.: MNHN 1974-93 (male, French Guyana), MNHN 1971-285 (female, French Guyana).
6. Ref.: MNHN 1884-2 (male, Mexico), KH 495 (unknown sex, Nicaragua).
7. Ref.: MNHN A-3324 (female, unknown origin), MNHN 1870-596 (female, unknown origin),

MNHN 1879-192 (unknown sex and origin),
MNHN 1913-71 (unknown sex and origin).

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